Lecture 2: Network flows

Dr. George Mertzios

george.mertzios@durham.ac.uk

Room 2066, MCS Building
Tel: 42429

Flow networks, flow, maximum flow

A common scenario:

- material is transfered in a network from a "source" to a "sink"
- source produces material at a steady rate, sink consumes at same rate

Examples:

- water through system of pipes
- current through electrical network
- data through network
- trucks from factory to warehouse

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Edges have given capacity:

- gallons per minute / trucks per hour / kB per second ...

Vertices (other than source/sink) are "junctions":

- material flows through them without collecting in them
- entering rate = exiting rate (flow conservation)

Some definitions

Maximum-flow problem:

- we wish to compute greatest possible rate of transportation from source to sink

Flow network:

- $G = (V, E)$ is directed graph
- two distinguished vertices: source s and sink t
- each edge $(u, v) \in E$ has non-negative capacity $c(u, v) \geq 0$
- if $(u, v) \notin E$, we assume $c(u, v) = 0$.

We assume that every vertex lies on some path from s to t:

- for each $v \in V$, there is a path $s \rightsquigarrow v \rightsquigarrow t$

$\Rightarrow$ G is connected $\Rightarrow$ $|E| \geq |V| - 1$

A flow in G is a real-valued function $f : V \times V \to \mathbb{R}$ that satisfies these three properties:

Definition (Flow constraints)

- Capacity constraint:
  For all $u, v \in V$, we require $f(u, v) \leq c(u, v)$
  Flow from one vertex to another must not exceed given capacity.

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- Flow conservation:
  For all $u \in V - \{s, t\}$, we require

$$\sum_{v \in V} f(u, v) = 0$$

  Total flow out of a vertex (other than source or sink) is 0.
  Can rewrite as: $\forall v \in V - \{s, t\} : \sum_{u \in V} f(u, v) = 0$,
  i.e. total flow into a vertex is 0.

$f(u, v)$ — can be positive, zero, or negative — is called flow from u to v

Definition (Total positive flow)

The **total positive flow entering** vertex **v** is

$$\sum_{\mathbf{u} \in V:\ f(u,v)>0} f(u, v)$$

Also, **total positive flow leaving** vertex **u** is

$$\sum_{\mathbf{v} \in V:\ f(u,v)>0} f(u, v)$$

Definition (Total net flow)

The **total net flow** at a vertex v is

$$\begin{aligned} & \text{total positive flow leaving } v \\ - \quad & \text{total positive flow entering } v \end{aligned}$$

Definition (Flow value)

The **value** of flow f is defined as the total flow leaving the source (and thus entering the sink):

$$|f| = \sum_{v \in V} f(s, v)$$

Note: $|\cdot|$ does not mean "absolute value" or "cardinality".

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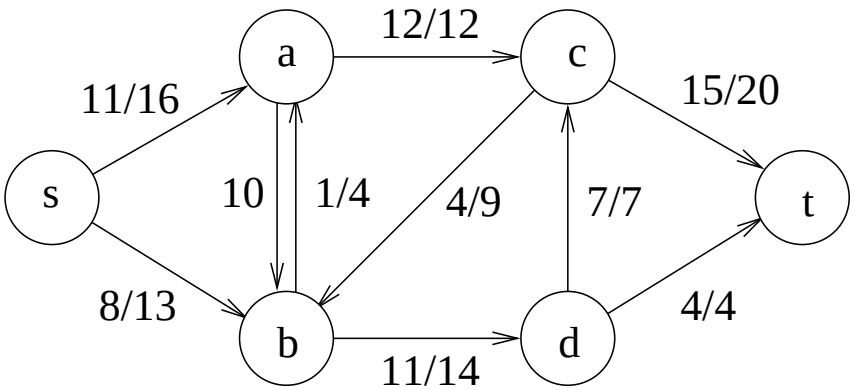


Single numbers on the edges denote edge capacities.

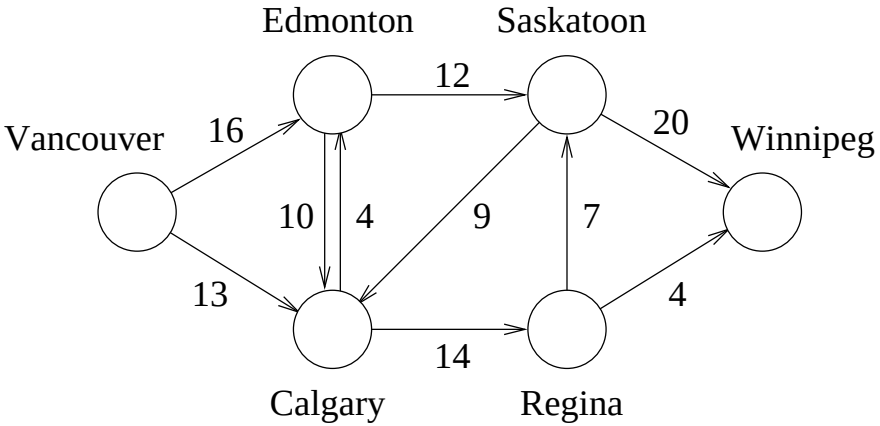
Pairs x/y denote $f(u, v)/c(u, v)$ (only labeled when $f(u, v) > 0$; if $f(u, v) \leq 0$, then only capacity is shown).

Figure shows flow f in G with value $|f| = 19$.

An example

The **Lucky Puck Company** produces hockey pucks:

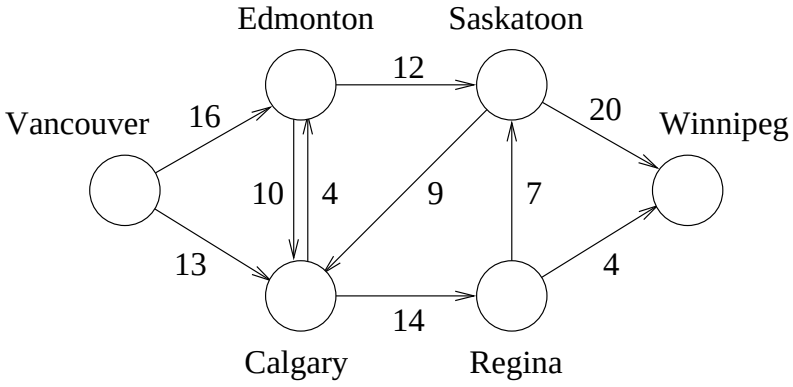
- it has factory (source) in Vancouver and warehouse (sink) in Winnipeg
- they lease space on trucks to ship pucks from factory to warehouse
- trucks have limited capacity
- trucks take specified routes (edges) between cities
- they can ship $\leq c(u, v)$ items per day from city u to city v
- goal is to find largest number p of items per day that can be shipped, and then to produce this amount each day
- no point in producing more pucks than what can be transported to warehouse



Seems appropriate to model "flow" of shipments with flow in network:

- have capacity constraint (items shipped per day from one city to another is limited)
- flow conservation must be obeyed (in steady state, rate at which pucks enter intermediate city must equal the rate at which they leave — otherwise, accumulation)

Cancellation



However, **one difference** between shipments and flows:

Suppose they ship 8 crates per day from Edmonton to Calgary, and 3 crates per day from Calgary to Edmonton.

Natural enough, but we can't represent these shipments directly by flows: **skew-symmetry is violated** [must have $f(u, v) = -f(v, u)$].

However, Lucky Puck's strategy is pointless anyway:

> why ship 8 crates from Edmonton to Calgary and 3 creates back, instead of just shipping 5 crates from Edmonton to Calgary?

In effect, 3 of the 8 crates from Edmonton to Calgary are **cancelled** by the 3 crates from Calgary to Edmonton.

We represent this with a flow: $f(Ed, Ca) = 5$ and $f(Ca, Ed) = -5$.

Technical tools

Implicit summation. Let $X, Y \subseteq V$. Then

$$f(X, Y) = \sum_{x \in X} \sum_{y \in Y} f(x, y)$$

For example, flow-conservation

$$\sum_{v \in V} f(u, v) = 0 \quad \forall u \in V - \{s, t\}$$

can be written simply as

$$f(u, V) = 0 \quad \forall u \in V - \{s, t\}$$

Technical tools (continued)

Commonly occurring identities:

1. For all $X \subseteq V$, we have $f(X, X) = 0$
   Because each $f(u, v)$ and $f(v, u) = -f(u, v)$ cancel each other.

2. For all $X, Y \subseteq V$, we have $f(X, Y) = -f(Y, X)$
   Generalization of $f(X, X) = 0$, same reasoning.

3. For all $X, Y, Z \subseteq V$ with $X \cap Y = \emptyset$, we have

$$f(X \cup Y, Z) = f(X, Z) + f(Y, Z)$$

$$f(Z, X \cup Y) = f(Z, X) + f(Z, Y)$$

   Split summation into two: one over X, one over Y.

The Ford-Fulkerson method

Three important ideas:

1. residual networks
2. augmenting paths
3. cuts

Method is iterative:

1. Start with $f(u, v) = 0$ for all $u, v \in V$
2. At each iteration, increase flow value by finding an augmenting path — a path from source to sink along which we can increase flow — and then augment flow along this path.
3. Repeat until no augmenting path can be found.

Residual networks

Idea: Residual network consists of edges that can admit more flow.

Formally: Consider vertices u and v. Amount of additional flow we can push from u to v before exceeding capacity $c(u, v)$ is residual capacity of (u, v):

$$c_f(u, v) = c(u, v) - f(u, v)$$

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Note: When flow $f(u, v)$ is negative, then residual capacity $c_f(u, v)$ is greater than $c(u, v)$.

Interpretation:

- flow of $-x$ from u to v (i.e. $f(u, v) = -x < 0$)
- implies flow of x from v to u (i.e. $f(v, u) = x > 0$)
- can be canceled by pushing x units from u back to v
- can then push another $c(u, v)$ from u to v

$\Rightarrow$ we can push in total $c_f(u, v) = c(u, v) + x > c(u, v)$ from u to v

Residual networks

Given flow network $G = (V, E)$ and flow f, the residual network of G induced by f is $G_f = (V, E_f)$ with

$$E_f = \{(u, v) \in V \times V : \; c_f(u, v) > 0\}$$

i.e. each residual edge that can admit flow that is strictly positive.

$$G: \quad u \underset{1}{\overset{3/5}{\rightleftarrows}} v \qquad \Longrightarrow \qquad G_f: \quad u \underset{4}{\overset{2}{\rightleftarrows}} v$$

Note: $|E_f| \leq 2|E|$.

Lemma

Let $G = (V, E)$ be a flow network, f be a flow in G, G_f be the residual network of G induced by f, and let f' be a flow in G_f. Then the flow sum $f + f'$ with

$$(f + f')(u, v) = f(u, v) + f'(u, v)$$

is a flow in G with value $|f + f'| = |f| + |f'|$.

Proof. Must verify that skew symmetry, capacity constraints, and conservation are obeyed.

Skew symmetry:

$$\begin{aligned}
(f + f')(u, v) &= f(u, v) + f'(u, v) \\
&= -f(v, u) - f'(v, u) \\
&= -(f(v, u) + f'(v, u)) \\
&= -(f + f')(v, u)
\end{aligned}$$

Capacity constraint: Note f' is flow in G_f, so $f'(u,v) \leq c_f(u,v)$. Since by def. $c_f(u,v) = c(u,v) - f(u,v)$,

$$\begin{aligned}
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&= c(u,v)
\end{aligned}$$

Flow conservation: For all $u \in V - \{s,t\}$,

$$\begin{aligned}
\sum_{v \in V} (f+f')(u,v) &= \sum_{v \in V} (f(u,v) + f'(u,v)) \\
&= \sum_{v \in V} f(u,v) + \sum_{v \in V} f'(u,v) \\
&= 0 + 0 \\
&= 0
\end{aligned}$$

Finally, value:

$$
\begin{aligned}
|f + f'| &= \sum_{v \in V} (f + f')(s, v) \\
&= \sum_{v \in V} (f(s, v) + f'(s, v)) \\
&= \sum_{v \in V} f(s, v) + \sum_{v \in V} f'(s, v) \\
&= |f| + |f'|
\end{aligned}
$$

$\square$

Augmenting paths

Given flow network $G = (V, E)$ and flow f:

- an augmenting path P is a simple path in residual network G_f

Recall: each edge (u, v) in G_f admits some additional positive flow, obeying capacity constraint.

Flow value can be increased by

$$c_f(P) = \min_{(u,v) \in P} c_f(u, v)$$

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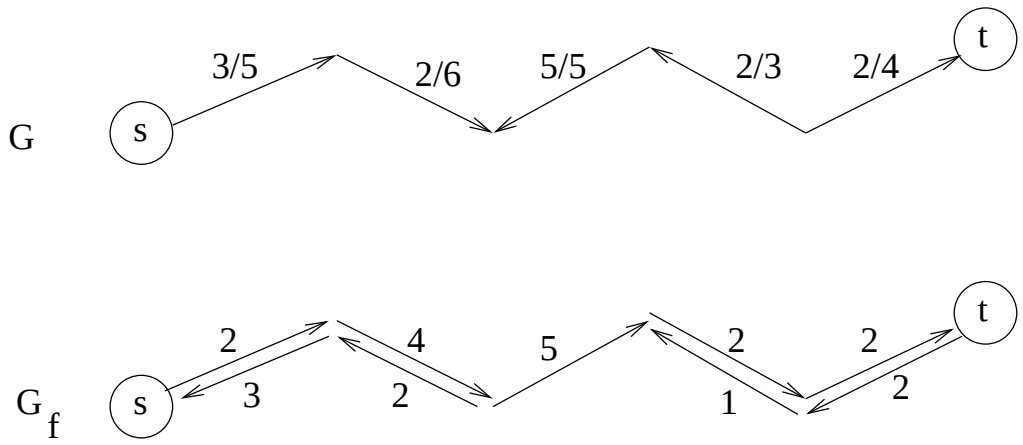

In this example: possible augmentation by at most ???

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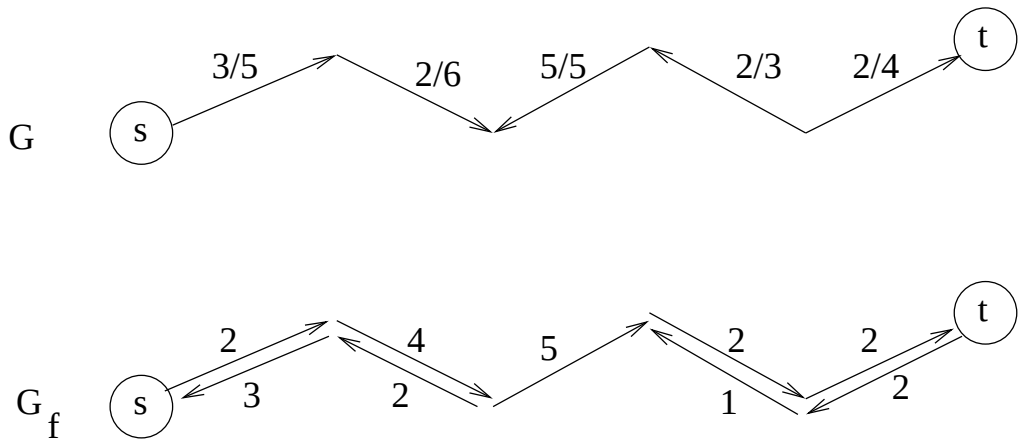

In this example: possible augmentation by at most 2

Lemma

Let $G = (V, E)$ be a flow network, f be a flow in G, and let P be an augmenting path in G_f. Define f_P by:

$$f_P(u,v) = \begin{cases} c_f(P) & \text{if } (u,v) \text{ is on } P \\ -c_f(P) & \text{if } (v,u) \text{ is on } P \\ 0 & \text{otherwise} \end{cases}$$

Then f_P is a flow in G_f with value $|f_P| = c_f(P) > 0$.

Proof: Exercise.

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Proof: Exercise.

Corollary (Improve)

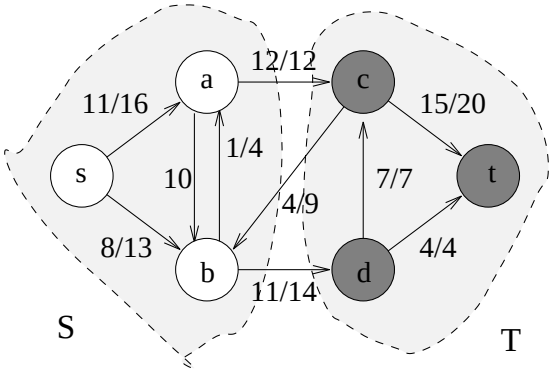
Let G, f, P, f_P be as above. Define $f' = f + f_P$. Then f' is a flow in G with value $|f'| = |f| + |f_P| > |f|$.

Proof: From last two lemmas.

Cuts of flow networks

A cut of a flow network $G = (V, E)$ is a partition of V into the sets S and $T = V - S$ such that $s \in S$ and $t \in T$.

If f is a flow in G, then $f(S, T)$ is the **net flow** across cut (S, T); its **capacity** is $c(S, T)$.



Flow: $f(S, T) = f(a, c) + f(b, c) + f(b, d) = 12 + (-4) + 11 = 19$
Note: can include negative flows.

Capacity: $c(S, T) = c(a, c) + c(b, d) = 12 + 14 = 26$
Note: only non-negative values (i.e. only edges from S to T).

A minimum cut is a cut with minimum capacity over all cuts.

Net flow across any cut is the same:

Lemma (Net flow)

Let f be a flow in a flow network G with source s and sink t, let (S, T) be a cut of G. Then the flow across (S, T) is $f(S, T) = |f|$.

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Corollary (Upper cut)

The value of a flow f in a network G is upper-bounded by the capacity of any cut (S, T) in G.

Proof.

$$|f| = f(S, T) = \sum_{u \in S} \sum_{v \in T} f(u, v) \leq \sum_{u \in S} \sum_{v \in T} c(u, v) = c(S, T)$$

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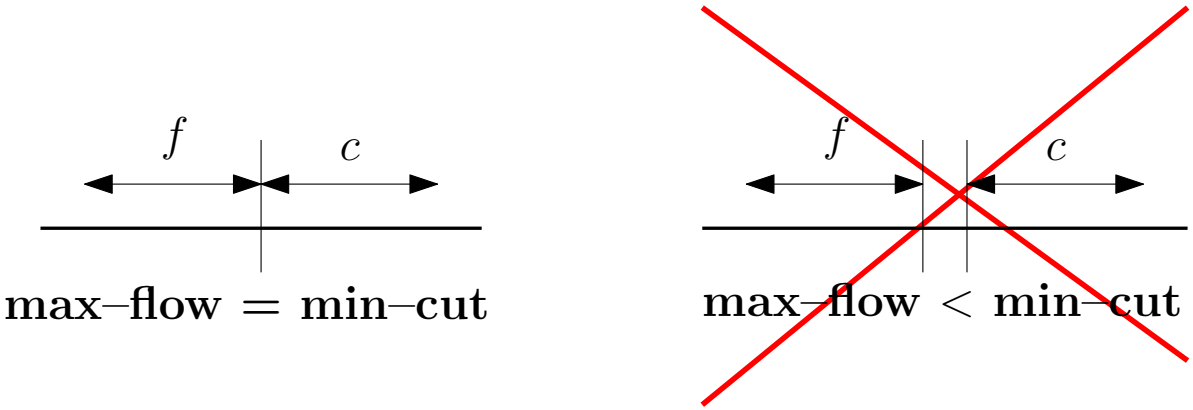
Therefore: **max–flow** $\leq$ **min–cut**

Improving last corollary, the next theorem shows equality:

Theorem (Max-flow min-cut)

If f is a flow in a flow network $G = (V, E)$ with source s and sink t, then the following conditions are equivalent:

1. *f is a maximum flow in G.*
2. *The residual network G_f contains no augmenting path.*
3. *$|f| = c(S, T)$ for some cut (S, T) of G.*



Basic Ford-Fulkerson algorithm

Input: G, s, t

1: **for** each edge $(u, v) \in E$ **do**
2: $\quad f(u, v) \leftarrow 0$
3: $\quad f(v, u) \leftarrow 0$
4: **end for**
5: **while** there exists a path $P = s \rightsquigarrow t$ in the residual network G_f **do** {i.e. augmenting path in G}
6: $\quad c_f(P) \leftarrow \min\{c_f(u, v) : (u, v) \in P\}$
7: $\quad$ **for** each edge $(u, v) \in P$ **do**
8: $\quad\quad f(u, v) \leftarrow f(u, v) + c_f(P)$
9: $\quad\quad f(v, u) \leftarrow -f(u, v)$
10: $\quad$ **end for**
11: **end while**

Question: How to find an augmenting path P of G in line 5?

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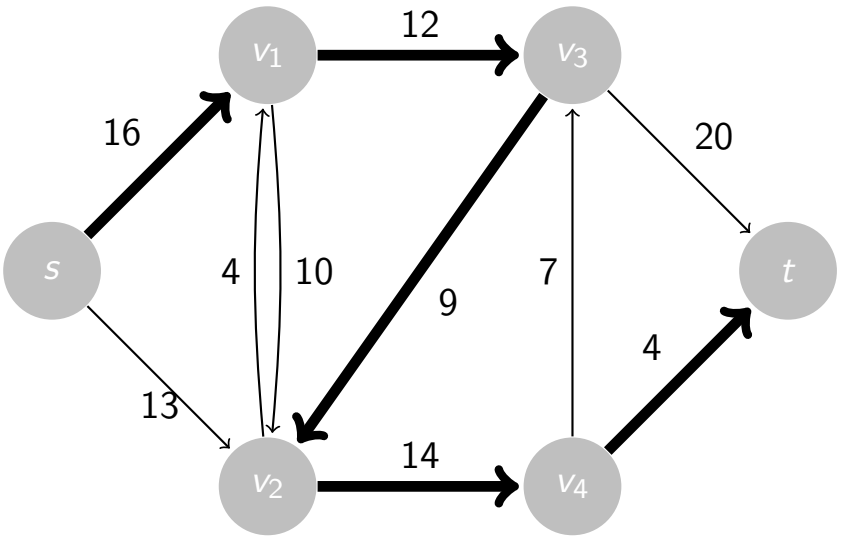
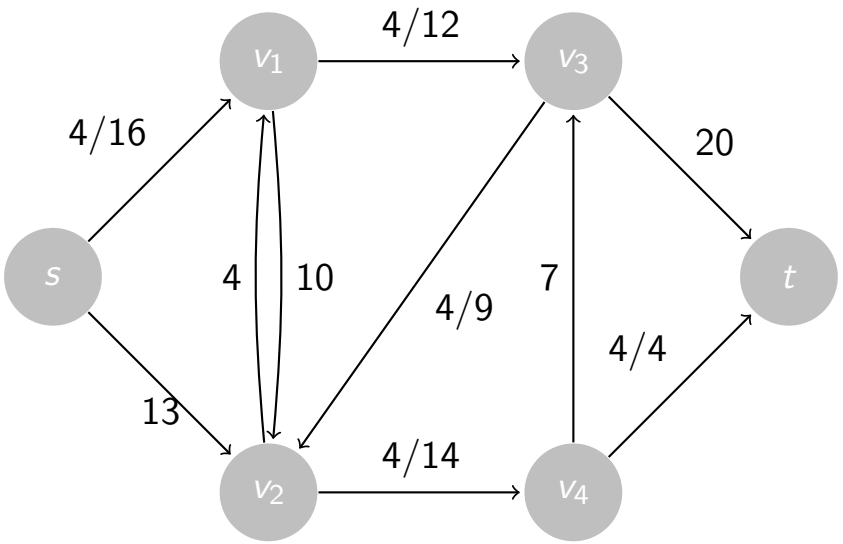
Question: How to find an augmenting path P of G in line 5?

Answer: Not specified by the algorithm !

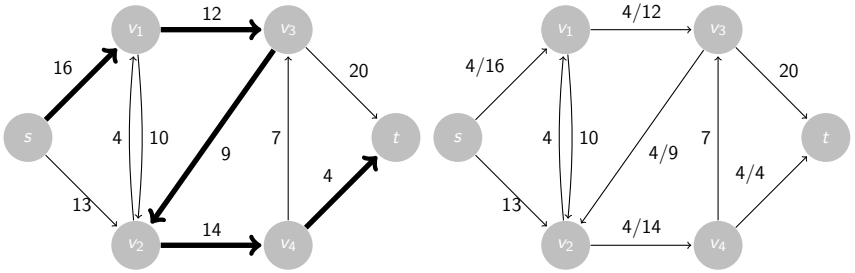
Example: step 1

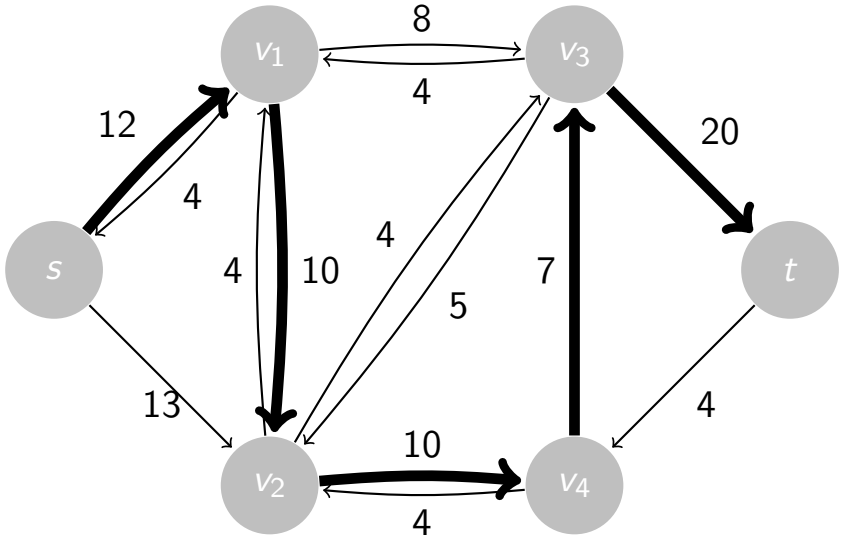


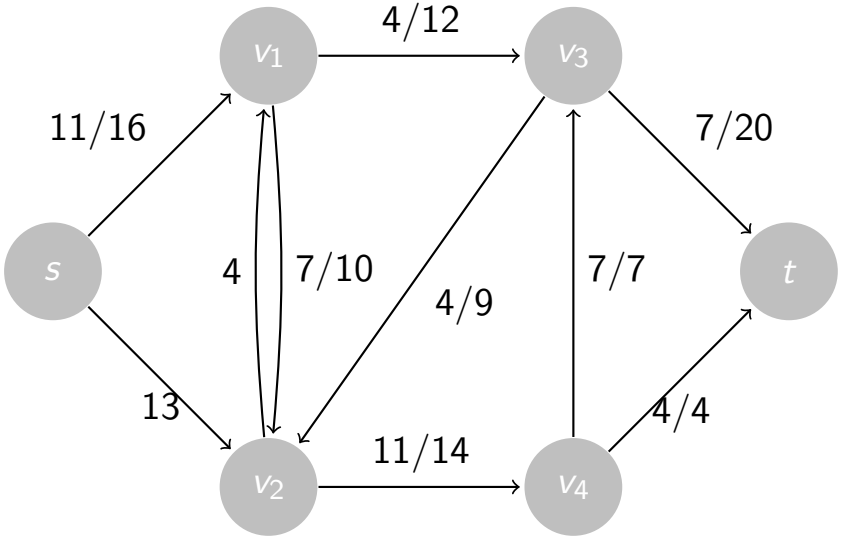


residual network on the left, resulting flow network on the right

Example: step 2







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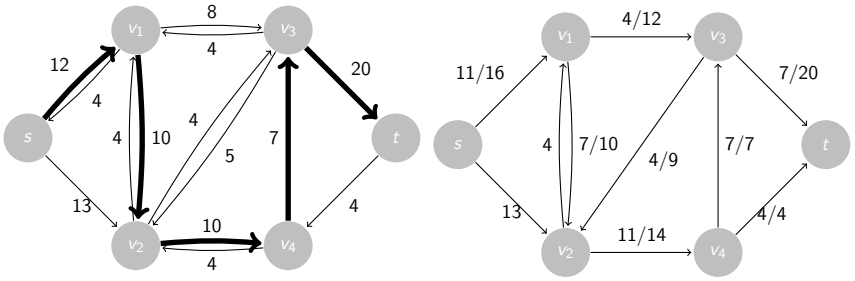
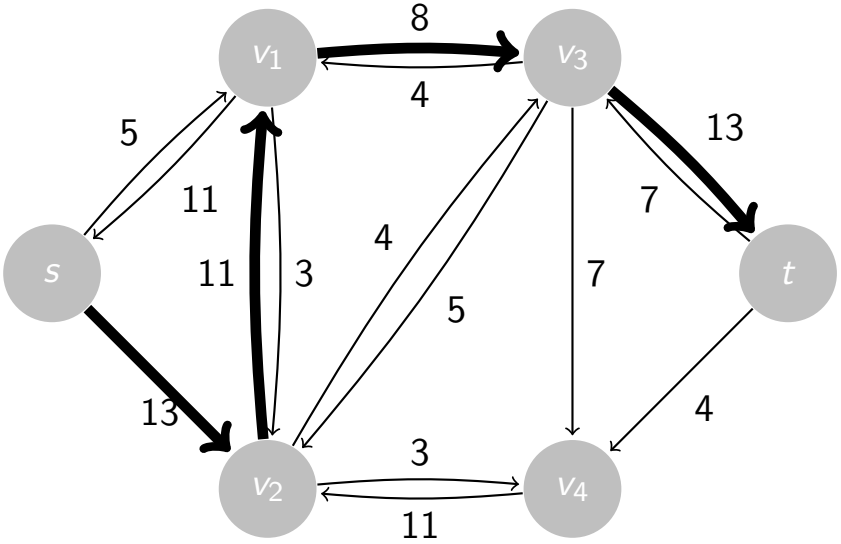
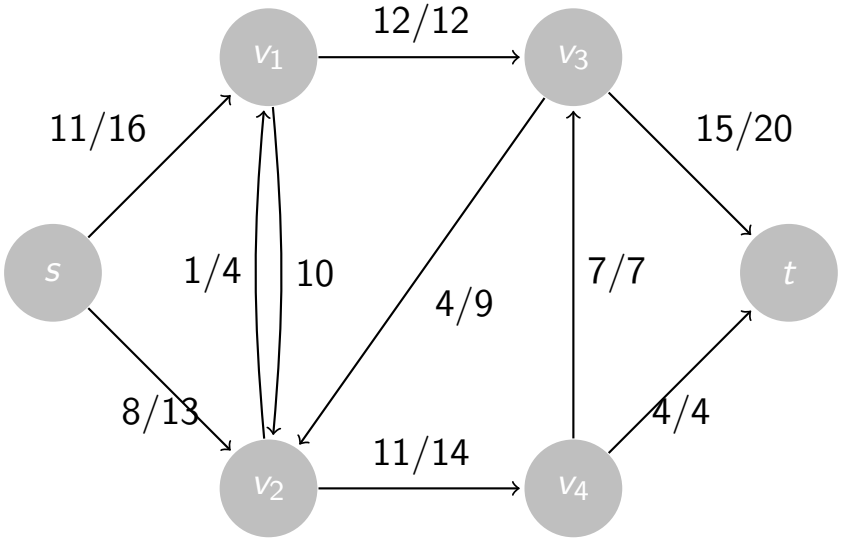
Example: step 3







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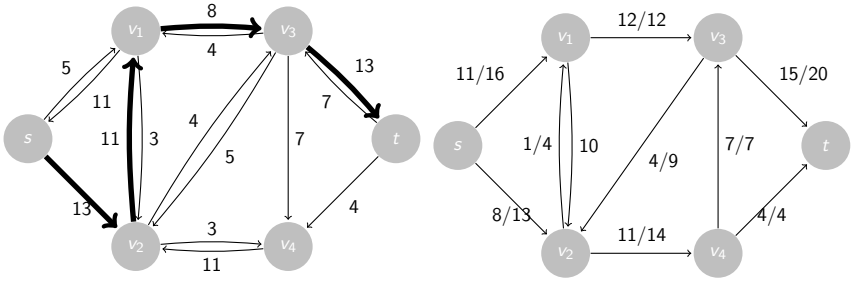
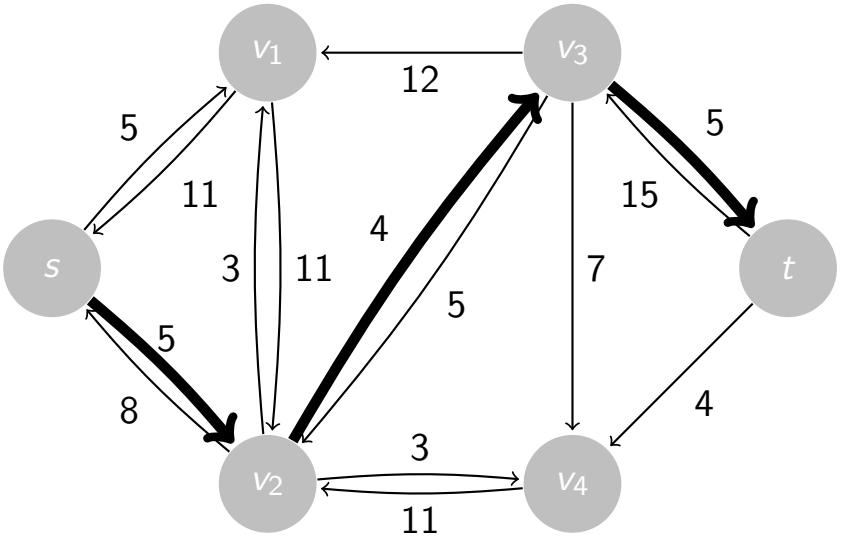
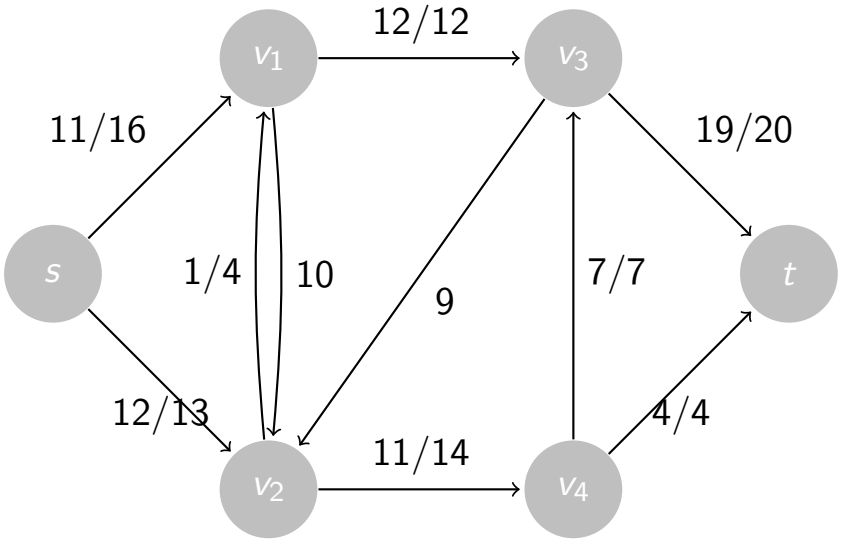
Example: step 4

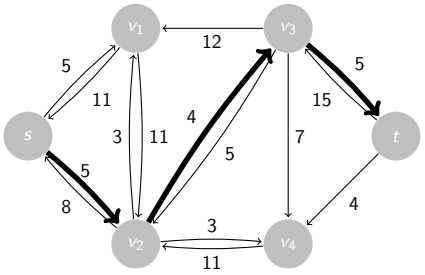


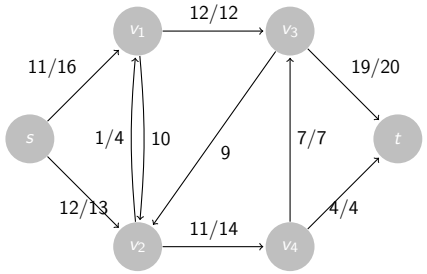


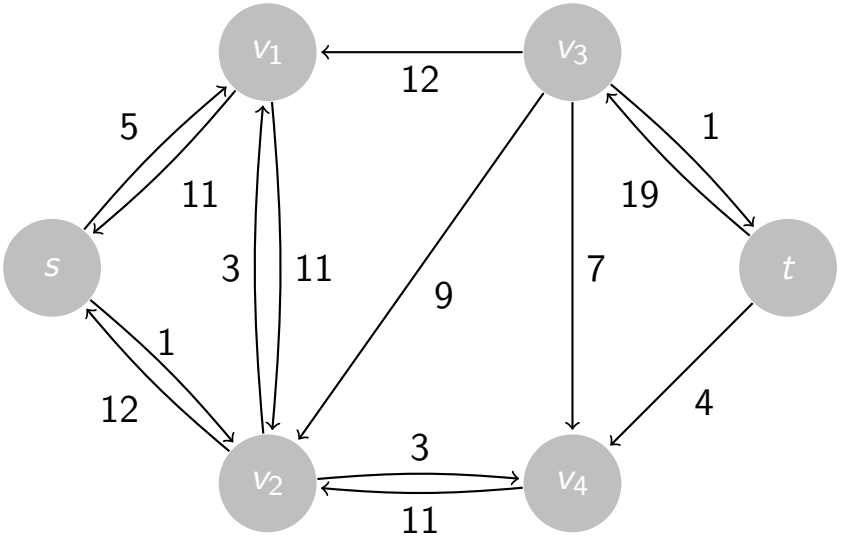


residual network on the left, resulting flow network on the right

Example: step 5







residual network; no $s - t$ path left $\Longrightarrow$ done!

Running time

It strongly depends on how augmenting paths in line 5 are determined.

If chosen poorly:

- value of flow increases with each iteration
- but possibly too slowly !
- in extreme cases it might never terminate,
  and it can even not converge to the value of maximum flow!
  (this can happen only if the capacities are irrational numbers)

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In practice:

- capacities are integers

$\Rightarrow$ always terminates (each time increase flow value by ≥ 1)

- if all capacities are small, then efficient algorithm (polynomial time), otherwise...

A simple running time analysis

Assume **integral capacities**. Simple bound $O(E \cdot |f^*|)$ for running time when choosing paths **arbitrarily**, $|f^*|$ being the **value of maximum flow**:

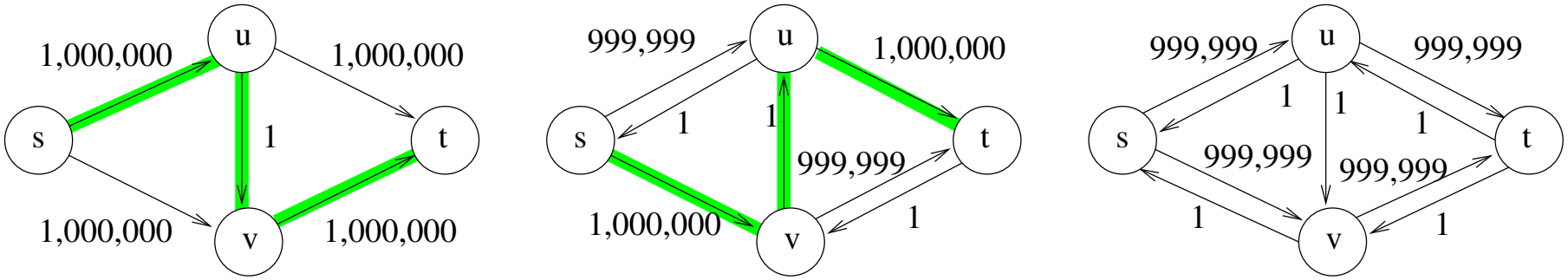
- **initialization** in lines 1–4 take $O(E)$
- **while** loop in lines 5–11 is executed at most $|f^*|$ times (value of flow increases by at least one unit in each iteration)
- in lines 6–10 (within the while loop) we need $O(V + E) = O(E)$ time (just graph traversing by DFS/BFS and then path traversing)

Input: G, s, t

1: **for** each edge $(u, v) \in E$ **do**
2: $\quad f(u, v) \leftarrow 0$
3: $\quad f(v, u) \leftarrow 0$
4: **end for**
5: **while** there exists a path $P = s \rightsquigarrow t$ in the residual network G_f **do** {i.e. augmenting path in G}
6: $\quad c_f(P) \leftarrow \min\{c_f(u, v) : (u, v) \in P\}$
7: $\quad$ **for** each edge $(u, v) \in P$ **do**
8: $\quad\quad f(u, v) \leftarrow f(u, v) + c_f(P)$
9: $\quad\quad f(v, u) \leftarrow -f(u, v)$
10: $\quad$ **end for**
11: **end while**

Problematic case:

Algorithm needs $\Omega(E|f^*|)$ time in worst case, because (u, v) is always chosen to be part of augmenting path. Here: $|f^*| = 2,000,000$



Important:

- the value of the maximum flow f^* can be arbitrarily large (might be even exponential on the size of the network!)

$\Rightarrow$ this algorithm has not polynomial running time in worst case

A polynomial algorithm

Edmonds-Karp algorithm:

- a special implementation of Ford-Fulkerson algorithm
- the augmenting path P is always chosen to be a shortest path from s to t (e.g. using BFS) in the residual network G_f
- regardless of the flow that fits in this path P

Running time:

- in total at most $O(V \cdot E)$ augmentations

$\Rightarrow$ running time $O(V \cdot E^2)$, since each augmentation needs $O(E)$ time (i.e. polynomial on the network size)

Edmonds-Karp algorithm: running time analysis

Lemma (Monotonicity of Edmonds-Karp algorithm)

For every vertex $v \in V - \{s, t\}$, the shortest-path distance $\delta_f(s, v)$ in the residual network G_f increases monotonically after each flow augmentation.

Proof idea.

Assume otherwise and derive a contradiction. □

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Proof idea.

Assume otherwise and derive a contradiction. □

Lemma (Number of flow augmentations)

The algorithm performs at most $O(V \cdot E)$ flow augmentations.

Proof idea.

- Every augmentation has at least one "critical" edge.
- Every edge becomes at most $\frac{V}{2}$ times "critical" during the algorithm.
- there are $O(E)$ edges $\Rightarrow$ at most $O(V \cdot E)$ augmentations. □

Applications of the Maximum Flow problem

The Dancing Party problem:

- a group X of men and a group Y of women, all in a dancing party
- $x \in X$ is "matchable" with $y \in Y$ if they are both willing to dance with each other
- what is the largest set of pairs happily dancing at the same time?

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Let $G = (V, E)$ be an undirected graph. A matching in G is a set of edges of E with no common endpoints.

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Maximum Matching Problem

Let $G = (V, E)$ be an undirected graph. A matching in G is a set of edges of E with no common endpoints.

- We "pair" neighboring vertices of V such that each vertex is "paired" with at most one other vertex of V.
- Special case: G is bipartite, i.e. V is partitioned into two sets X, Y such that every edge is between one $x \in X$ and one $y \in Y$.

Maximum Bipartite Matching

Many further applications, for example:

- X is a set of machines and Y a set of jobs
- $x \in X$ is connected $y \in Y$ if job y can be executed on machine x
- maximum matching $\Leftrightarrow$ maximum number of executed jobs

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We can simply solve this problem using Maximum Flows:

- let $G = (X, Y, E)$ be a bipartite graph
- we construct from G a flow network H such that every maximum flow in H can be immediately converted to a maximum matching in G

$\rightarrow$ then apply Ford-Fulkerson algorithm on H !

That is: we "reduce" the bipartite matching problem to the maximum flow problem (which we know how to solve)

Maximum Bipartite Matching

1. we include in H all vertices of G, plus a new source s and sink t
2. we add all edges of G and direct them all from X to Y

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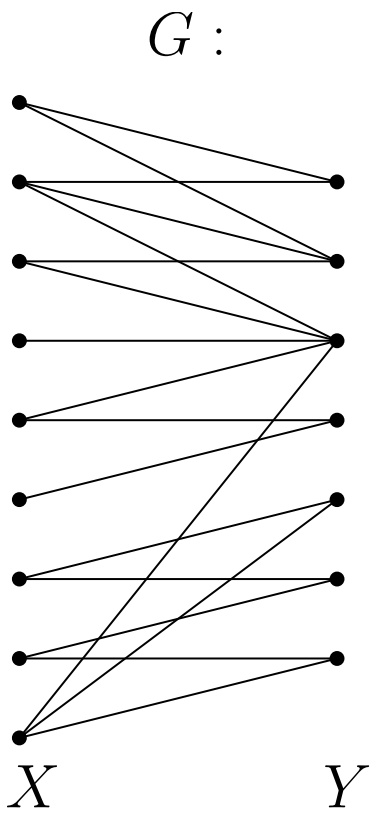

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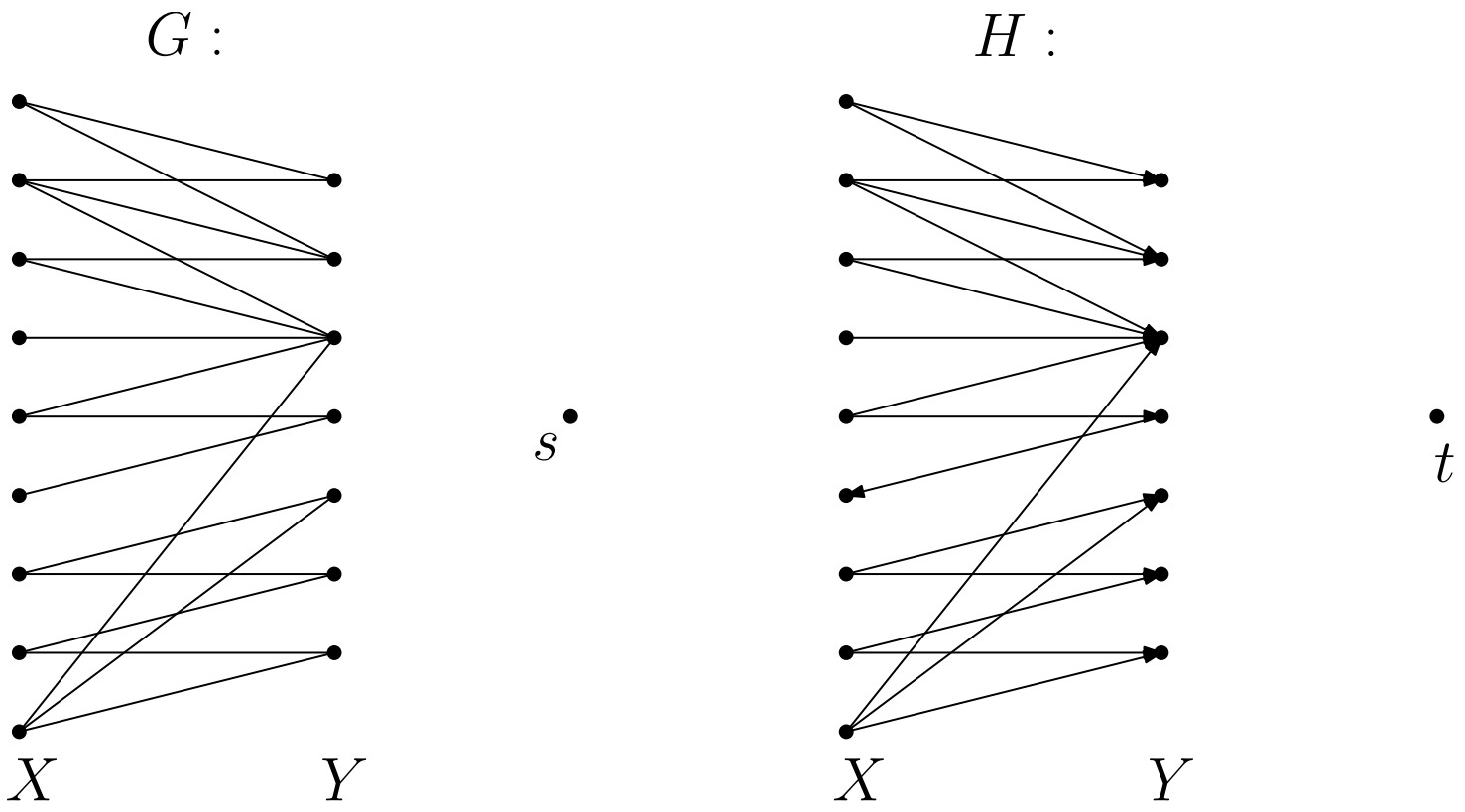

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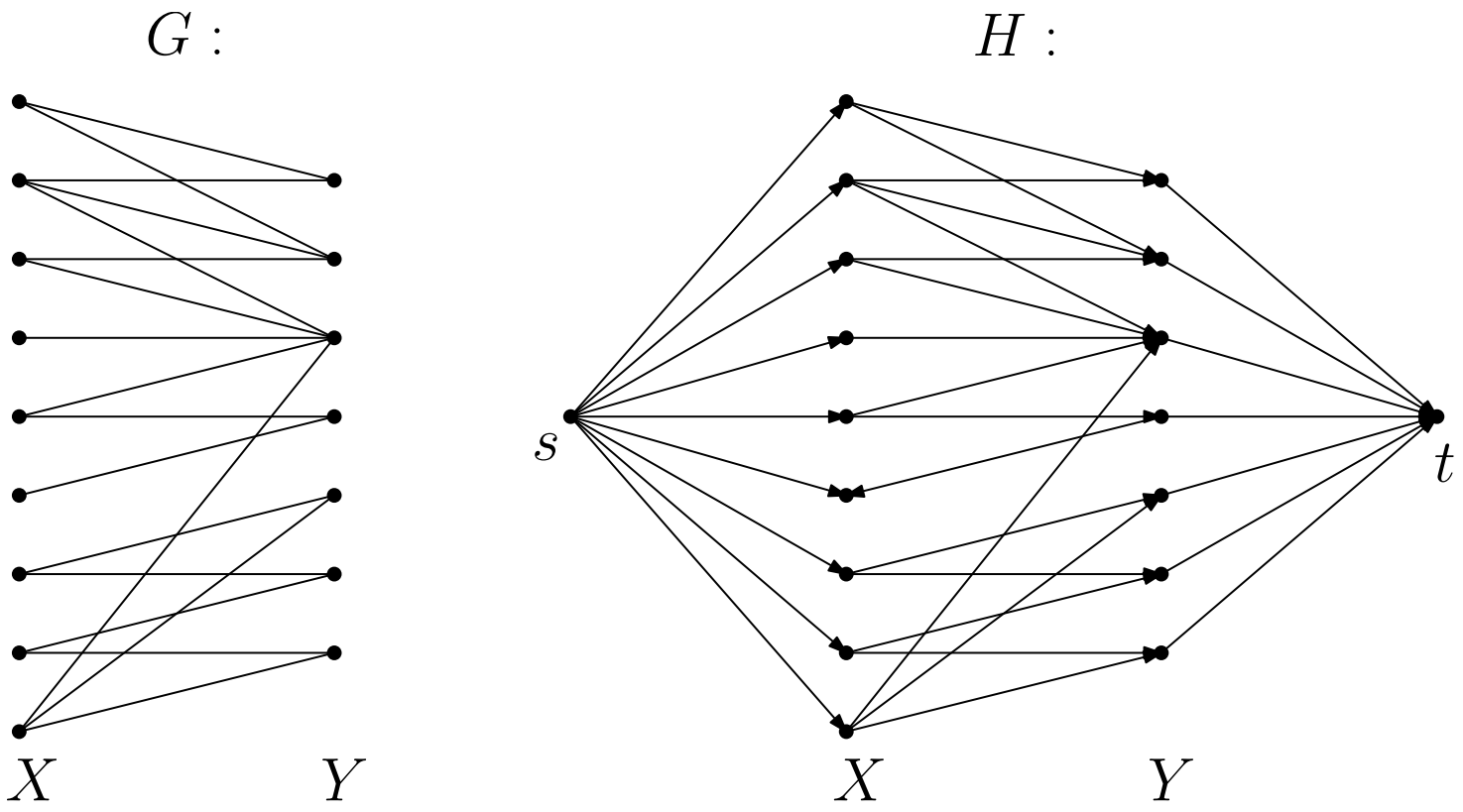

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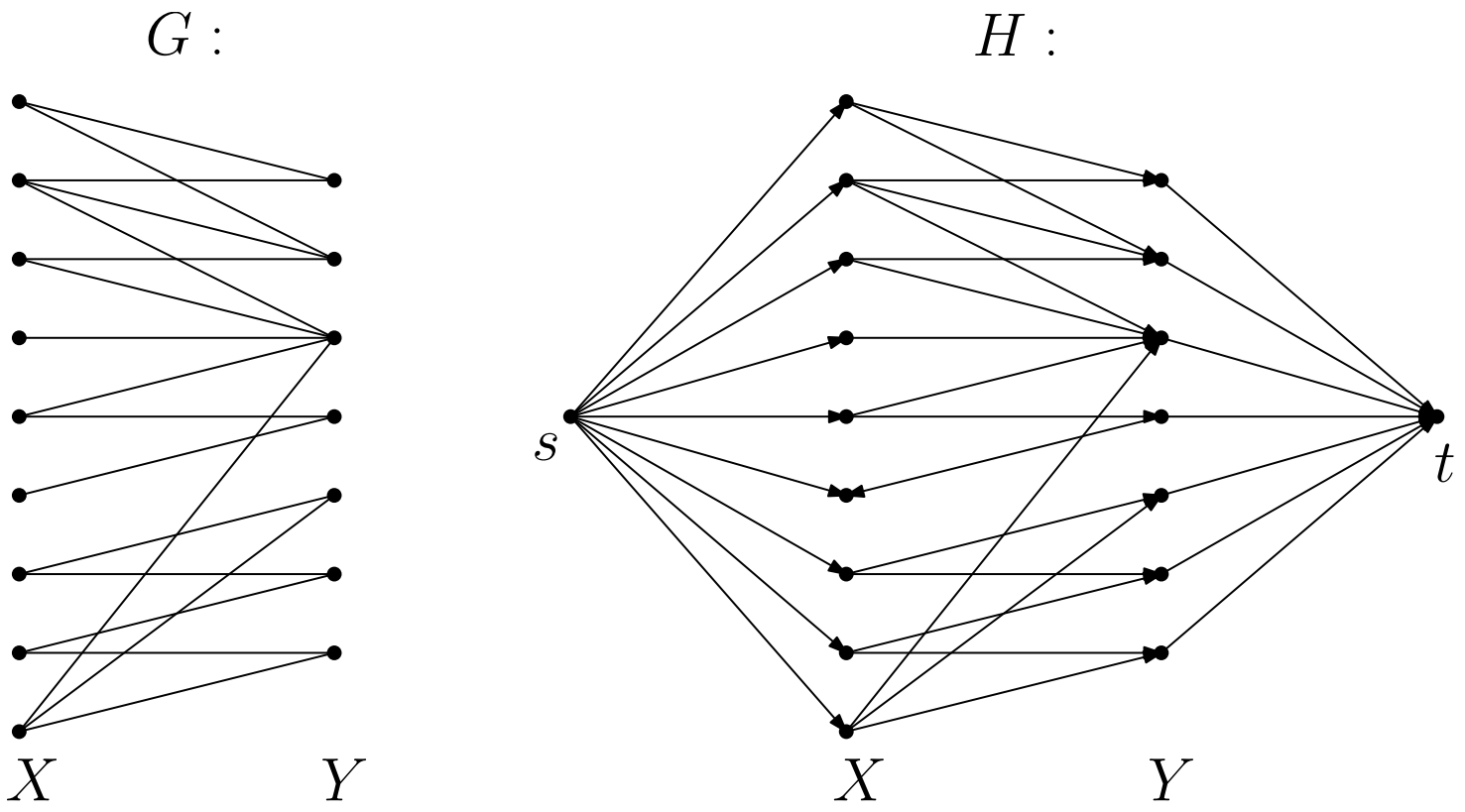

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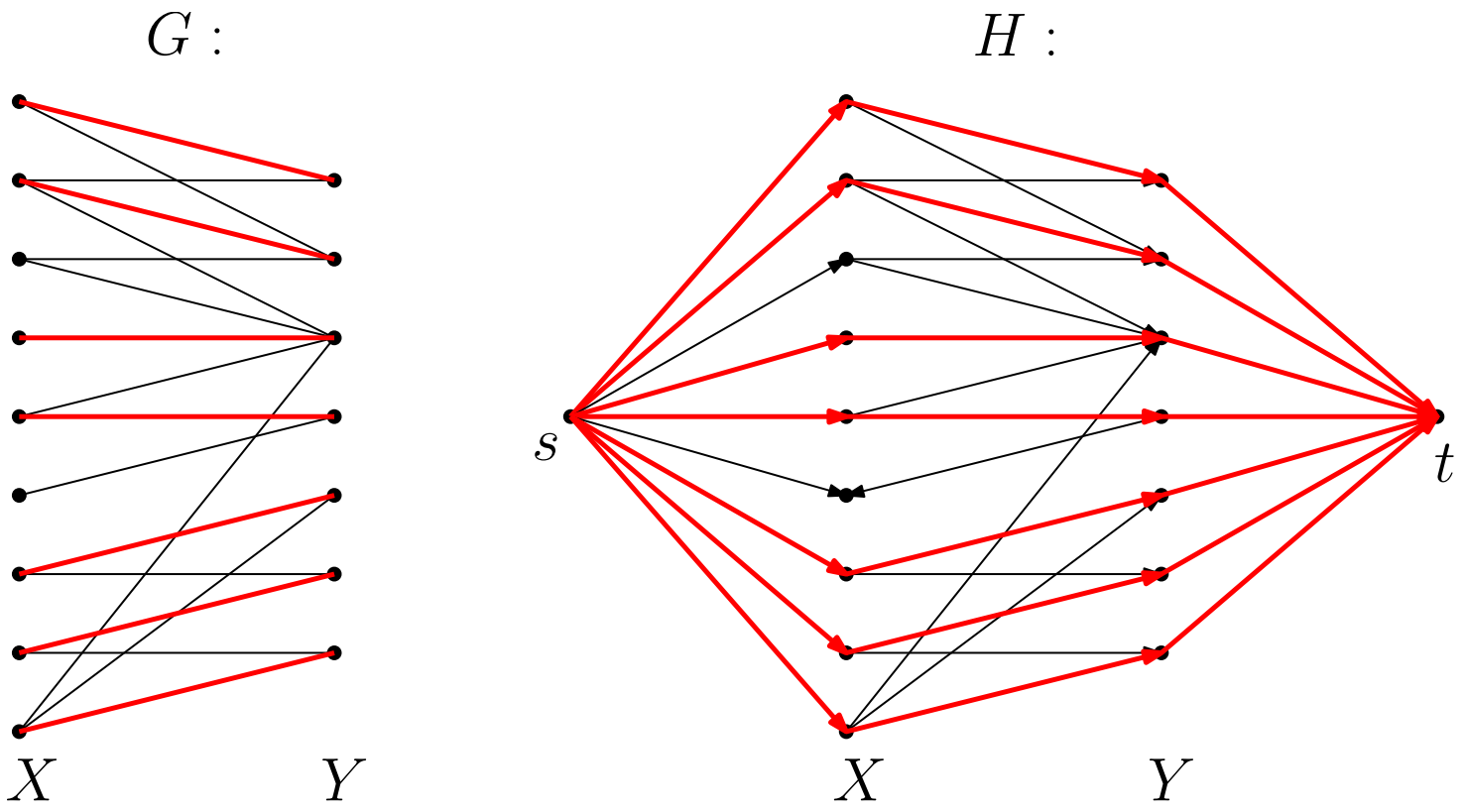

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$\Rightarrow$ each vertex in X is paired by M with at most one vertex of Y and vice versa

$\Rightarrow$ M is a matching, where $|M| = |f|$

Maximum Bipartite Matching

Conversely, given a maximum matching M in G:

1. for each edge e of H that also belongs to G:
   - if $e \in M$, we set $f(e) = 1$,
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Therefore: maximum matching in G = maximum flow in H

Maximum Bipartite Matching: running time

Given a bipartite graph G with n vertices and m edges:

- we construct H from G in $O(n+m)$ time
- we compute a maximum flow in H using (the standard) Ford-Fulkerson algorithm
- the value of the maximum flow is $|f| = |M| \leq \frac{n}{2}$

$\Rightarrow$ at most $\frac{n}{2}$ flow augmentations

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The fastest known algorithm has running time: $O(m \cdot \sqrt{n})$
(by simultaneously augmenting many paths)

Additional slides

Proof of the "Net flow" lemma.

Note $V = T \cup S$.

$$\begin{aligned} f(S,T) &= f(S,V) - f(S,S) = f(S,V) - 0 \\ &= f(s,V) + f(S-s,V) = f(s,V) + 0 \quad \text{(why???)} \\ &= |f| \text{ by def.} \end{aligned}$$

$\square$

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$$S = \{v \in V : \exists \text{ path from } s \text{ to } v \text{ in } G_f\} \quad \text{and} \quad T = V - S$$

Partition (S, T) is a cut because $s \in S$ and $t \notin S$. For each $u \in S, v \in T$ we have $f(u, v) = c(u, v)$, because otherwise $(u, v) \in E_f$, v would be in S ($s \rightsquigarrow u \to v$). By earlier Lemma "Net flow", $|f| = f(S, T) = c(S, T)$.

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(3) $\Longrightarrow$ (1) [$|f| = c(S, T)$ for some cut $(S, T) \Rightarrow f$ max flow]
By earlier corollary "Upper cut", $|f| \leq c(S, T)$ for all cuts (S, T).
Condition $|f| = c(S, T)$ implies that f is maximum flow. $\square$